

This is the Team Round. You have 20 minutes to complete 10 problems.

This is a team round.

There is no penalty for guessing. Answer submissions are only valid if they are written on the answer sheet below.

Team name			
p	Answer	p	Answer
1		6	
2		7	
3		8	
4		9	
5		10	

Problem 1. How many integers x satisfy $|x - 1434| \leq 2025$?

Problem 2. Let x and y be natural numbers such that

$$\gcd(x, y) = 69 \quad \text{and} \quad \text{lcm}(x, y) = 207.$$

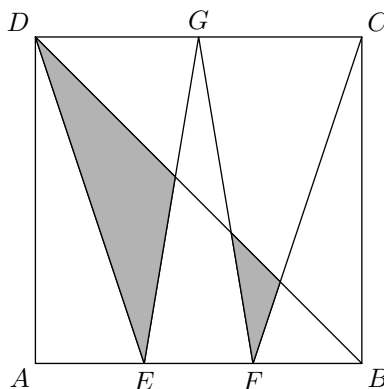
Determine xy .

Problem 3. An integer is *fiveminded* if it is divisible by 5 but does not contain the digit 5. How many three-digit fiveminded integers are there?

Problem 4. The polynomial $x^3 - 22x^2 + 27x + c$ has two roots which differ by exactly 2. What is the maximum possible value of c ?

Problem 5. Jeff writes down all of the permutations of the digits of 14341434 on the board. He calls a permutation *funny* if 434 is a substring of the permutation. (A substring must have all of its characters consecutive and in order). How many funny permutations did Jeff write on the whiteboard?

Problem 6. In square $ABCD$ below, $DG = GC$, and $AE = EF = FB$. What is the ratio of the shaded area to the area of the square?



Problem 7. Zack flips two unfair coins, each of which lands heads with probability $\frac{2}{3}$. Simultaneously, Jack flips two unfair coins, each of which lands heads with probability x , where $x > \frac{1}{2}$. A person wins if both of their coins land on the same face (either both heads or both tails), and the other person does not. If neither get matching faces on a flip, the process is repeated until someone does. If both get matching faces on the same flip, the game immediately ends and no one wins. Given that the probability that Jack wins is $\frac{2}{5}$, find the value of x .

Problem 8. The sum of the positive divisors of a number x , including 1 and x , is 248. Find the sum of all possible values of x .

Problem 9. In right triangle ABC , $AB = 24 + 6\sqrt{2}$, $AC = 24 - 6\sqrt{2}$, and $\angle BAC = 90^\circ$. Points D and E lie on segment AC such that $AD = DE = EC$ and D is between A and E , and points F and G lie on segment AB such that $AF = FG = GB$ and F lies between A and G . Let C' be the reflection of C across G . Line ℓ passes through C' and intersects rays $\overrightarrow{AB}$ and $\overrightarrow{AC}$ at P and Q , respectively. If $\angle QPA = \angle FDA$, what is the value of $C'P \cdot C'Q$?

Problem 10. For all positive integers n , denote

$$f(n) = \frac{2n+1}{(n^2+2n+1)\sqrt[3]{n^2} + (n^2+n)\sqrt[3]{n^2+n} + n^2\sqrt[3]{n^2+2n+1}}.$$

Compute $f(1) + \cdots + f(999)$.